

Volatility-Regime Identification and Forecasting in TSMC

Using Gaussian Hidden Markov Models

Thomas Gerrard MPhys Physics with Theoretical Physics University of Bath

August 4, 2026

Abstract

Financial-market volatility evolves through distinct and persistent regimes, meaning that observations recorded on adjacent trading days may be generated under materially different market conditions. This study applies a Gaussian Hidden Markov Model to volatility-related features for Taiwan Semiconductor Manufacturing Company (TSM), using data from January 2024 to December 2025 for estimation and observations from 2026 for out-of-sample evaluation. Four economically interpretable regimes were identified, comprising low-volatility bull, momentum-bull, high-volatility and heavy-tailed risk states. The regimes displayed strong temporal persistence, with estimated self-transition probabilities reaching approximately 0.97, and retained broadly consistent economic characteristics out of sample. High-volatility observations exhibited average daily realised volatility 0.955 percentage points above that of the low-volatility bull regime. Regime-conditioned EWMA and GARCH forecasts were evaluated against their corresponding general specifications. The general EWMA produced the strongest out-of-sample performance, with RMSE, MAE and QLIKE values of 0.253, 0.195 and 2.833, respectively. Both forecasting comparisons indicated that incorporating the decoded regime directly into the variance recursion did not improve point-forecast accuracy. The results therefore support the HMM primarily as a tool for market-state identification and risk interpretation rather than as a mechanism for automatically improving short-horizon volatility forecasts.

Contents

1	Introduction	4
1.1	Motivation	4
1.2	Research Objectives	4
1.3	Report Structure	5
2	Literature Review	5
2.1	Hidden Markov Models and Market Regimes	5
2.2	Alternate Volatility Modelling Techniques	6
3	Data	8
3.1	Asset Selection	8
3.2	Data Sources	9
3.3	Feature Engineering	10
3.3.1	Log Returns	10
3.3.2	Rolling Skew	11
3.3.3	Rolling Realised Volatility	11
3.3.4	Drawdown	12
3.3.5	VIX Level	13
3.3.6	Trading Volume	13
3.3.7	Volatility Ratio	13
4	Methodology	14
4.1	Model Specification	14
4.2	Parameter Estimation	14
4.3	Model Selection	16
4.4	Regime Inference	17
4.5	Out-of-Sample Validation	18
5	Results	18
5.1	Identified Regimes	19
5.2	Regime Characteristics	22
5.3	Train-Test Comparison	25
6	Conditional Volatility Estimation	26
6.1	Regime-Conditioned Volatility Modelling - EWMA	27
6.2	Regime-Conditioned Volatility Modelling - GARCH (1,1)	29
7	Extensions	32
7.1	Heavy Tailed Emission Distributions	32
7.2	Dynamical Hidden Markov Models	32
7.3	Extended Testing	33
8	Conclusion	33

8.1	Summary of Findings	34
8.2	Project Limitations	34
8.3	Future Work	35
References		36

1 Introduction

Volatility plays a central role in financial markets. It affects portfolio risk, derivative pricing and market stability. Understanding how volatility evolves over time is therefore of considerable importance to both investors and researchers. Financial markets demonstrate periods of distinctly different behaviour and even this behaviour has been known to develop over time. Extended periods of stability are interrupted by high volatility regimes or durations of large price movements given the effects of macroeconomic events and news. This behaviour has suggested evidence for latent market regimes.

1.1 Motivation

Hidden Markov Models provide a natural framework for modelling regime-dependent behaviour by allowing market dynamics to be governed by an unobservable hidden state process. By considering the current observational information up to a time t , the HMM considers the most likely path taken to reach the state we are currently in, further proposing a likelihood of future states, conditional on our current state and our available information.

The motivation is to investigate whether Hidden Markov Models can identify meaningful volatility regimes in TSMC stock and whether these regimes enhance the accuracy of volatility forecasting frameworks.

1.2 Research Objectives

The first objective of this study is to develop and implement a Gaussian Hidden Markov Model using different sets of volatility-related features. The most effective model was concluded by comparing AIC, BIC and log likelihood scores across all of the models trained.

Secondly, the identified set of candidate regimes were tested to determine whether these states correspond to interpretable and meaningful market states. This involves analysing statistical characteristics over the possible states, to establish if the model successfully distinguishes between different environments.

Following this, the third objective is to assess the robustness of the inferred states through out-of-sample validation. This requires determining whether the regime structure remains economically interpretable and statistically stable when the fitted HMM is applied to unseen data. Evaluating consistency in regime characteristics, persistence and occupancy across the training and test samples provides evidence on whether the model generalises beyond the estimation period.

Finally, this report examined whether the volatility regimes identified by the Hidden Markov Model contain useful information about future realised volatility. Regime-dependent forecasts were compared with simpler benchmark models to assess whether conditioning on the prevailing market state improved short-horizon forecasting performance. This provides a direct evaluation of the practical value of regime classification and helps distinguish between the HMM's usefulness as a tool for identifying changing market conditions and its ability to enhance quantitative conditional volatility estimates.

1.3 Report Structure

Chapter 2 provides an insight into the relevant literature regarding the applications of volatility regimes and Hidden Markov Models to quantitative finance.

Chapter 3 and 4 describes the data and methodology employed throughout the study. This includes the selection of market data, the construction of volatility-related features and the implementation of the Hidden Markov Model framework. The model selection process and validation methodology are also discussed.

Chapter 5 provides an overview of the market regimes identified by the Hidden Markov Model. The regime characteristics were analysed and significant conclusions drawn from them, using both financial and statistical metrics. Particular importance is placed on the stability of regimes out-of-sample and multi stock.

Chapter 6 investigates the forecasting implications of the inferred regimes, with state-dependent volatility forecasting frameworks, EWMA and GARCH compared to a rolling realised volatility benchmark and their own respective general models.

2 Literature Review

The purpose of this literature review is to establish the theoretical and practical foundations upon which the subsequent modelling framework is developed. Rather than providing an exhaustive survey of Hidden Markov Models or financial time series analysis, this chapter critically examines the academic literature that has most strongly influenced the conception and evolution of this project. The discussion begins by introducing the concept of market regimes and the motivation for modelling financial markets as systems characterised by distinct latent states. This is followed by a review of volatility modelling frameworks before delving into the specifics of Hidden Markov Models and their application to regime identification within quantitative finance.

The review concludes by extending the discussion beyond historical volatility towards implied volatility and the options market. Although these topics lie outside the primary scope of the present study, they provide a natural progression of the ideas developed throughout the report and illustrate potential applications of regime detection to derivative pricing, volatility forecasting and options trading strategies. Consequently, this chapter not only motivates the methodological choices adopted in the remainder of the dissertation but also highlights opportunities for future development and extension of the proposed modelling framework.

2.1 Hidden Markov Models and Market Regimes

Hidden Markov Models (HMM) offer an alternative method to approaching non-stationarity within financial markets. Assuming these nonlinear dynamics arise due to shifts in market regime we can deploy

HMM to help decode these different market states and begin to explain behaviour. The financial markets give us noisy observations for which there are underlying processes we cannot observe, and these processes are generated by hidden states that evolve over time according to a Markov process.

Rabiner identifies three central problems associated with Hidden Markov Models (Rabiner 1989). A method must be proposed to calculate the probability of a sequence of outputs when parameters are given. Secondly, the inferred regimes need decoding, which Rabiner suggests can be done by considering the most likely sequence of latent states according to current observations. Thirdly, the most likely set of state transitions and output probabilities need computing when the sequence of output is provided. Methods in which the Forward-Backward Algorithm solve the first problem, the Viterbi algorithm decoded regimes, and the Baum-Welch (EM) algorithm can be used to compute the most likely set of state transitions, are all methods discussed by Rabiner.

Furthermore, Ntantamis extends the conventional Hidden Markov Model to a Duration Hidden Markov Model (DHMM), in which the duration of each hidden state is modelled explicitly (Ntantamis 2010). Although the methodology is closely related to the framework adopted in this report, several differences are important. Ntantamis employs the Maximum Posterior Mode rule (MPM) rather than Viterbi decoding. Whereas the Viterbi algorithm identifies the most probable complete sequence of hidden states, MPM assigns the most probable state to each observation using the posterior state probabilities. A further distinction, and the principal contribution of the paper, lies in its treatment of regime duration. In this report, expected regime duration is implied by the estimated transition matrix and remains fixed for each state. By contrast, Ntantamis models duration as a time-varying stochastic quantity that depends on exogenous variables and must therefore be estimated directly,

$$\log(\tau) = \kappa_j^T Z + \zeta_j W, \quad (1)$$

where τ denotes the duration of a given regime and Z contains the exogenous variables included in the duration specification. Ntantamis uses short-term interest rates, the interest-rate spread and an intercept term, while κ_j is the state-specific coefficient vector that determines how these variables affect regime duration. The model therefore treats duration as a stochastic, time-varying quantity influenced by both macroeconomic conditions and random variation. This provides a more flexible representation of market persistence than the conventional HMM adopted in this report, where expected state duration is implied solely by the estimated transition probabilities.

2.2 Alternate Volatility Modelling Techniques

There exist many existing methodologies for modelling volatility, however, this section will focus majorly on the use of GARCH and stochastic volatility models. Generalised Autoregressive Conditional Heteroskedasticity models provide a natural benchmark for the volatility component of this study because they represent conditional variance as a recursive function of past return shocks and past variance. In

the standard GARCH(1,1) specification,

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (2)$$

where α measures the immediate response of volatility to new shocks and β captures the persistence of previous volatility. This structure is well suited to the empirical feature of volatility clustering, whereby periods of elevated volatility tend to be followed by further elevated volatility. Within this project, the GARCH(1,1) model is not used to infer market regimes directly. Instead, it provides a continuous conditional-volatility estimate against which the usefulness of the HMM regime structure can be assessed. A general GARCH model is fitted across the full return series, while separate state-dependent specifications are also considered to test whether conditioning on the inferred HMM regimes improves the representation of volatility dynamics. This creates a direct comparison between a conventional model in which volatility evolves continuously through past shocks and a regime-based framework in which broader market characteristics change through transitions between latent states. The results therefore allow the HMM to be evaluated not only in terms of economic interpretability and out-of-sample regime persistence, but also in terms of whether regime information adds value beyond a standard GARCH benchmark.

Although extensions such as GJR-GARCH and long-memory specifications can account for asymmetry and slower-decaying persistence (Ding 2016), the GARCH(1,1) model is retained here because it offers a parsimonious and widely used benchmark that remains directly comparable with the state-dependent models estimated later in the report.

Bergomi (2015) offers a separate view, in which he proposes that volatility remains a continuous process that evolves as stochastic diffusion. The concept that variance follows its own stochastic differential equation (SDE) is the central idea of the Heston model, given by

$$dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t}dW_t^v, \quad (3)$$

this equation encoding terms of mean reversion and long-run variance. Bergomi's criticism is that directly modelling v_t is not the most natural object and instead he proposes to model forward variance, considering the idea that options markets form a forward variance curve as they observe future implied variances as

$$\xi_t(T) = \mathbf{E}_t[v_t]. \quad (4)$$

Rather than one variance, there exists an entire function that Bergomi leverages to model

$$d\xi_t(T) = \omega_t(T)dW_t. \quad (5)$$

Stochastic-volatility models reproduce features such as volatility smiles, skew, term structures and the leverage effect, which are not captured by the Black–Scholes framework. Bergomi develops both one-factor and two-factor specifications, with the latter providing a richer representation of volatility-surface dynamics and the behaviour of the VIX. The framework also links forward variance to instantaneous variance, thereby maintaining consistency with broader stochastic-volatility modelling approaches (Bergomi

2015).

GARCH models provide a recursive framework for modelling conditional volatility from past shocks and previous variance, whereas Bergomi's framework treats future variance as a forward-looking stochastic process. The present study is positioned between these approaches by using a Gaussian Hidden Markov Model to identify discrete latent market regimes from observed return- and volatility-related features, while employing GARCH(1,1) as a complementary conditional-volatility benchmark. This allows the analysis to assess whether regime information provides explanatory value beyond a conventional continuous volatility model.

This study is positioned between the discrete regime-switching literature and the continuous conditional-volatility literature. Unlike GARCH models, which represent volatility through a recursive function of past shocks and conditional variance, the Gaussian Hidden Markov Model identifies a finite set of latent market environments from a multivariate collection of return-, volatility- and market-risk-related observations. The model therefore seeks to capture broader structural changes in market behaviour rather than volatility dynamics alone. At the same time, general and state-dependent GARCH specifications are estimated as complementary conditional-volatility benchmarks, allowing the usefulness of regime conditioning to be assessed against a conventional time-series framework. The analysis further extends beyond in-sample regime identification by evaluating whether the inferred economic characteristics and transition structure remain stable over a chronologically separated test period. Finally, comparison with options-implied volatility provides an exploratory link between historically inferred market regimes and the forward-looking expectations embedded in derivative prices. In this way, the study contributes a TSMC-specific empirical framework combining multivariate regime detection, out-of-sample validation and conditional-volatility comparison, while recognising that a complete integration of regime dynamics with the forward-variance models described by Bergomi remains an avenue for future research.

3 Data

The performance of a Hidden Markov Model is fundamentally dependent upon the quality and informational content of its observed feature vector. Consequently, before model specification and parameter estimation can be undertaken, it is necessary to define the market data from which the observations are constructed and justify the selection of each explanatory variable. This section outlines the financial asset selected for analysis, the corresponding data sources, and the feature engineering procedures used to transform the raw market data into a multivariate time series suitable for Gaussian Hidden Markov Model estimation. Particular emphasis is placed on ensuring that the resulting feature set captures complementary aspects of market behaviour, including returns, volatility and downside risk, whilst remaining computationally tractable for regime identification.

3.1 Asset Selection

This study focuses on Taiwan Semiconductor Manufacturing Company Limited, whose American Depositary Shares trade on the New York Stock Exchange under the ticker TSM. TSMC provides a suitable case study because its share price has experienced both comparatively stable and highly volatile periods over the sample examined. The company occupies a central position within the global semiconductor

supply chain as a leading pure-play foundry (Taiwan Semiconductor Manufacturing Company Limited 2026) and a major provider of advanced process and packaging technologies used in high-performance computing and artificial-intelligence applications. TSMC reports that demand for leading-edge silicon and advanced packaging has been supported by rapid growth in AI-related computing, while its technologies are used in products including AI accelerators, graphics processors and data-centre hardware.

TSMC is also one of the world's largest publicly traded semiconductor companies. Its market capitalisation was approximately \$1.88 trillion in late July 2026 (Stock Analysis 2026), although this value varies with the share price and exchange rate. Its scale, technological importance and exposure to global semiconductor demand mean that its valuation may respond to firm-specific developments, changes in AI-related demand, supply-chain conditions and wider macroeconomic or geopolitical shocks.

The liquidity of TSM's American Depositary Shares further supports its use in volatility analysis. On 27 July 2026, approximately 17.9 million shares were traded, while the security also had a broad listed options chain containing observable prices, implied volatilities, trading volumes and open interest across strikes and maturities. This provides sufficient market activity for analysing both return-based conditional volatility and option-implied volatility. More generally, market-microstructure research shows that bid-ask bounce and other liquidity effects can bias volatility estimates constructed from transaction prices. Selecting a liquid security therefore reduces, although does not eliminate, concerns that measured volatility is being driven by trading frictions rather than changes in the underlying market process. (Han et al. 2015)

Finally, the availability of a sufficiently long daily history for prices, returns, trading volume and market-risk variables permits the construction of rolling volatility features and the chronological separation of training and test samples required by the Hidden Markov Model. The combination of economic significance, changing volatility conditions, market liquidity and data availability therefore makes TSM an appropriate asset through which to evaluate regime identification and state-dependent conditional-volatility estimates.

3.2 Data Sources

Our primary source of data is Yahoo Finance, a free API, from which we first gathered daily data for TSM spanning 01/01/24 to 30/06/2026. The available downloaded variables include OHCL data, alongside traded volume and the datetime. We imported both TSM and VIX data separately and held these in respective dataframes as they required index alignment. The timezones for both dataframes were removed, allowing us to align them and merge them to form one complete dataframe. Furthermore, we created other columns in our dataframe with features that will be soon discussed before cleaning our data by dropping all NaN values. There did not exist many NaN values so this is okay as we keep most of our data and there are no big missing gaps across the dataframe that would lead to significant result inaccuracy when evaluating the results of the HMM. Given our timeframe of data, the chosen train test split was approximately 75/25, with the train sample spanning the 01/01/2024 to 31/12/2025, and the test sample covered all of 2026 (~ 6 months). Ordering our train-test split chronologically allows us to conserve time-series nature of the data and avoid look-ahead bias when training the model.

3.3 Feature Engineering

The predictive capability of any machine learning model is fundamentally dependent upon the quality and relevance of the information provided as input. In the context of Hidden Markov Models, the observed feature vectors represent the only information available from which the latent market regimes can be inferred. Consequently, careful feature selection is essential to ensure that the model captures the characteristics associated with different volatility environments rather than simply describing price movements.

The features employed throughout this study were selected with the objective of characterising multiple aspects of market behaviour. Rather than relying solely on historical returns, the feature set incorporates measures of realised volatility, market-implied volatility, downside risk and trading activity. Together, these variables provide complementary information regarding both the current state of the market and the conditions that frequently accompany transitions between periods of low and high volatility.

Each feature was chosen based upon its economic interpretation and its expected contribution towards distinguishing between latent market regimes. Several engineered features were calculated using overlapping rolling windows and were therefore expected to exhibit serial dependence. For example, consecutive observations of 20-day realised volatility share 19 underlying daily returns. This creates smoother and more persistent feature series than the original daily data. Such persistence was considered when interpreting the duration and transition behaviour of the inferred regimes. The construction of each variable is described in the following subsections.

Table 1: Summary of volatility-related features used within the Hidden Markov Model.

Feature	Symbol	Purpose
Log Returns	r_t	Measure daily asset price movements and direction.
Rolling Skew	S_r	Captures the time-evolution of price asymmetry.
Rolling Realised Volatility	σ_{20}	Estimate recent realised market volatility.
Drawdown	DD_t	Quantify downside risk from previous price peaks.
VIX Level	VIX_t	Represent market-implied volatility expectations.
Trading Volume	V_t	Capture market participation and liquidity.
Volatility Ratio	VR	Encodes the relationship between short and long term volatility.

3.3.1 Log Returns

Daily log returns were selected as the primary measure of price movement throughout this study. In comparison to simple percentage returns, log returns possess several desirable statistical properties that make them particularly well suited to quantitative financial modelling. Most notably, they are additive through time, allowing multi-period returns to be obtained by summing individual daily observations. Furthermore, for the relatively small daily price changes observed in liquid equity markets, log returns closely approximate continuously compounded returns while remaining mathematically convenient for statistical analysis.

The daily log return was calculated using the closing price of the asset according to

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right), \quad (6)$$

where P_t denotes the closing price on trading day t .

Although the Hidden Markov Model developed in this study is primarily intended to identify latent volatility regimes, daily returns remain an essential component of the observation vector. Periods of elevated volatility are frequently accompanied by larger absolute price movements, whilst sustained bullish and bearish market environments often exhibit distinct return characteristics. Including log returns therefore enables the model to distinguish not only between differing levels of market uncertainty but also between regimes exhibiting contrasting price dynamics.

Finally, the use of log returns ensures consistency with much of the existing quantitative finance literature, where they form the standard representation of financial price changes and provide a natural foundation for subsequent volatility estimation.

3.3.2 Rolling Skew

Whilst volatility measures the dispersion of returns, skewness characterises the asymmetry of the return distribution. Rolling skewness was therefore included as an additional feature in order to capture changes in the shape of market returns that may not be reflected by volatility alone. Financial markets frequently exhibit asymmetric return behaviour, particularly during periods of market stress where large negative returns occur more frequently than large positive returns. Consequently, rolling skewness provides information regarding the balance of upside and downside price movements within a recent observation window.

Computing skewness on a rolling basis enables the Hidden Markov Model to observe how the asymmetry of the return distribution evolves over time. Persistent changes in rolling skewness may indicate transitions between different market environments, allowing the model to distinguish between regimes that exhibit similar levels of realised volatility but differing distributions of returns. In particular, periods of financial stress are often associated with increasingly negative skewness, reflecting a greater probability of large downside price movements.

Rolling skewness was calculated over a 20-day rolling window according to

$$S_r = \frac{1}{N} \sum_{i=1}^N \frac{(r_i - \mu)^3}{\sigma^3} \quad (7)$$

where S_t denotes the rolling skewness at time t , r_i represents the log returns contained within the rolling window, μ is the sample mean and σ is the sample standard deviation.

3.3.3 Rolling Realised Volatility

Rolling realised volatility was included to provide the Hidden Markov Model with a direct measure of the recent magnitude of price fluctuations. Unlike daily returns, which describe the direction and size of individual price movements, realised volatility characterises the variability of returns over a

specified historical window and therefore provides a more stable representation of the prevailing market environment.

Financial markets are well known to exhibit volatility clustering, whereby periods of elevated volatility tend to be followed by further periods of elevated volatility, whilst tranquil market conditions similarly display a high degree of persistence. A rolling estimate of realised volatility is therefore well suited to regime detection, as it captures these persistent changes in market behaviour whilst reducing the influence of isolated daily price movements that may not represent genuine changes in the underlying market state.

Throughout this study, realised volatility was estimated using a 20-day rolling window, corresponding approximately to one trading month. Whilst the rolling window inevitably smooths short-lived fluctuations in volatility, this behaviour is desirable for regime identification since latent market states are expected to persist over multiple trading sessions rather than arise from isolated observations. Consequently, only sustained changes in market behaviour produce substantial changes in the realised volatility estimate, allowing the Hidden Markov Model to identify persistent volatility regimes more reliably. These values are computed using

$$\sigma_{20} = \frac{1}{19} \sum_{i=1}^{19} (r_{t-i} - \bar{r}_t)^2 \quad (8)$$

where RV_{20} is the rolling volatility across a 20-day window and all other variables take previously defined forms.

3.3.4 Drawdown

Drawdown was included as an additional feature in order to quantify cumulative downside risk within the market. Whilst daily returns describe individual price movements, drawdown measures the percentage decline from the most recent historical price peak, thereby providing information regarding the magnitude and persistence of market declines. As such, it captures a different aspect of market behaviour from both returns and realised volatility.

Periods of substantial drawdown are commonly associated with episodes of financial stress, heightened uncertainty and prolonged bearish market conditions. Consequently, drawdown provides the Hidden Markov Model with information regarding the severity of sustained market declines, enabling it to distinguish between temporary periods of elevated volatility and more persistent market downturns. This distinction is particularly valuable when identifying latent volatility regimes, as markets experiencing similar levels of realised volatility may nevertheless exhibit markedly different downside risk profiles.

Throughout this study, drawdown was calculated relative to the cumulative maximum closing price observed up to each trading day, according to

$$DD_t = \frac{P_t - M_t}{M_t}, \quad (9)$$

where M_t denotes the cumulative maximum of the closing price observed up to trading day t , P_t again being our closing price for trading day t . Consequently, drawdown is always non-positive, taking a value of zero whenever the asset reaches a new historical high and becoming increasingly negative as the price declines from its most recent peak.

3.3.5 VIX Level

In addition to historical measures of market behaviour, the CBOE Volatility Index (VIX) was incorporated into the observation vector as a proxy for the market's expectation of future volatility. Unlike realised volatility, which is calculated from historical price movements, the VIX is derived from the prices of S&P 500 index options and reflects the level of implied volatility priced by market participants over the coming thirty calendar days. As such, it provides a forward-looking measure of market uncertainty.

Although the VIX is calculated using options on the S&P 500 rather than TSM specifically, movements in the index are widely regarded as representative of broader market sentiment and systemic risk. Periods of elevated market uncertainty are frequently accompanied by increases in the VIX, reflecting a greater demand for downside protection through the options market. Consequently, the inclusion of the VIX enables the Hidden Markov Model to incorporate information regarding prevailing market expectations alongside asset-specific historical observations.

3.3.6 Trading Volume

Daily trading volume was incorporated into the observation vector as a measure of market participation and information flow. Whilst price-based features describe how the market has moved, trading volume provides insight into the level of activity underlying those price movements. Periods of elevated trading volume frequently coincide with the release of new information, changes in investor sentiment and increased uncertainty, all of which may contribute to transitions between latent market regimes.

Unlike returns or realised volatility, trading volume does not directly quantify market risk. Instead, it acts as a complementary indicator of market behaviour by measuring the intensity of trading activity. Large increases in volume often accompany periods of heightened price discovery, whilst sustained periods of relatively low volume are commonly associated with more stable market conditions. Consequently, trading volume may assist the Hidden Markov Model in distinguishing between regimes that exhibit similar price dynamics but differing levels of market participation.

Throughout this study, the daily traded volume reported by Yahoo Finance was used directly without further transformation. Although volume is not itself a measure of volatility, its well-documented relationship with information arrival and market activity provides additional context that may improve the model's ability to identify persistent changes in market behaviour.

3.3.7 Volatility Ratio

Including a volatility ratio helps capture the relationship between short-term and long-term market variability. Whilst rolling realised volatility provides an estimate of the prevailing level of market uncertainty, it does not explicitly indicate whether volatility is increasing or decreasing relative to its longer-term trend. The volatility ratio therefore provides a measure of the current volatility environment relative to its historical baseline.

This feature is particularly relevant for regime detection, as transitions between latent market states are frequently accompanied by changes in the persistence and intensity of volatility. A volatility ratio greater than unity indicates that recent realised volatility exceeds its longer-term average, suggesting that

market uncertainty is increasing and that the asset may be transitioning into a higher-volatility regime. Conversely, values below unity indicate that recent volatility is relatively subdued compared with its historical behaviour, which is more commonly associated with stable market conditions.

The ratio is computed as

$$VR = \frac{\sigma_{10}}{\sigma_{60}}, \quad (10)$$

using the rolling volatilities of both a 10 day and 60 day window.

4 Methodology

This chapter presents the complete modelling framework used to identify latent volatility regimes in TSM. It begins by specifying the Gaussian Hidden Markov Model and outlining the estimation of its parameters, before comparing alternative model configurations and selecting the preferred specification. The chapter then describes the procedures used to infer the most likely regime sequence and assess whether the resulting regime structure remains economically meaningful and statistically stable out of sample.

4.1 Model Specification

The Hidden Markov Model implemented in this study assumes that the observed feature vectors are generated from multivariate Gaussian emission distributions, resulting in a Gaussian Hidden Markov Model (GHMM). Each hidden state is associated with its own multivariate Gaussian emission distribution, parameterised by a state-specific mean vector and covariance matrix where $o_t | x_t = i \sim N(\mu_i, \Sigma_i)$. These parameters determine the expected behaviour and variability of the observed feature vector when the market occupies a given latent regime. The hidden market regimes are assumed to evolve according to a first-order Markov chain, therefore exerting a first-order Markov property. An important modelling assumption is that the transition matrix for a Gaussian HMM is assumed constant, a true limitation of the model, as this considers states follow a time-homogeneous Markov chain. However, this is still an effective simple model for investigative purposes, as introducing a time-dependence on the transition matrix increases complexity; the Forward-Backward algorithms becomes more complicated and closed form parameter updates become less likely. Although computationally convenient, the assumption of a constant transition matrix may be restrictive when applied to financial markets, where macroeconomic conditions and market sentiment for example, can also influence regime-switching behaviour over time. Furthermore, this key assumption does not introduce look-forward bias as model parameters are estimated solely on the training sample before validation using out-of-sample testing.

Despite these limitations, the Gaussian Hidden Markov Model provides an efficient and highly interpretable framework for the exploratory analysis of latent market regimes. Further financial indicators can be suggested and investigated without the addition of unnecessary model complexity.

4.2 Parameter Estimation

Initially, the parameter set is unknown, denoted as $\theta = (\pi, A, \mu, \Sigma)$. Using the HMM framework, the objective is to estimate these parameters from the observed feature vectors by maximising the likelihood

of the observed data. The obvious difficulty lies in the 'hidden' nature of the market states, meaning direct maximum likelihood estimation is not possible. As introduced before, the Baum-Welch algorithm is capable of computing this by iteratively estimating the hidden state probabilities and updating the model parameters to maximise the likelihood of the observed data, allowing the Hidden Markov Model to be trained despite the underlying market regimes being unobservable.

Posterior probabilities are soft-assigned to our observations, with the probability of state occupancy given by

$$\gamma_t(i) = \frac{\alpha_t(i)\beta_t(i)}{\sum_{k=1}^K \alpha_t(k)\beta_t(k)}, \quad (11)$$

where $\alpha_t(i)$ and $\beta_t(i)$ are the forward and backward probabilities respectively. Transition probabilities also exist, defined as, $\xi_t(i, j) = P(x_t = i, x_{t+1} = j | X, \theta^{old})$, which is calculated as

$$\xi_t(i, j) = \frac{\alpha_t(i)A_{ij}b_j(x_{t+1})\beta_{t+1}(j)}{\sum_{i=1, j=1}^K \alpha_t(i)A_{ij}b_j(x_{t+1})\beta_{t+1}(j)}, \quad (12)$$

where all variables take on previously defined definitions. The algorithm then uses these soft assignments to update the model parameters, denoted by θ , by maximising the weighted likelihood of the observed data. The new initial state distribution becomes $\pi_i^{new} = \gamma_1(i)$, meaning the new probability of a beginning state i equals the posterior probability that the first observation belongs to that state. Similarly, the parameters update after each interval; our new transition matrix is computed as

$$A_{ij}^{new} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)}, \quad (13)$$

with A_{ij}^{new} becoming the expected number of transitions from i to j divided by the expected number of times state i is occupied. The new mean vector for multivariate observations is calculated as

$$\mu_i^{new} = \frac{\sum_{t=1}^T \gamma_t(i) o_t}{\sum_{t=1}^T \gamma_t(i)}, \quad (14)$$

becoming the weighted average of all observations. Finally, the updated covariance matrix is given as

$$\Sigma_i^{new} = \frac{\sum_{t=1}^T \gamma_t(i) (o_t - \mu_i)(o_t - \mu_i)^T}{\sum_{t=1}^T \gamma_t(i)}, \quad (15)$$

which is the weighted covariance matrix of all observations assigned probabilistically to state i .

Iterating the process, the Baum-Welch algorithm continues to update the parameters and increase the likelihood that the latest parameter set is responsible for our observation data. The algorithm provides a final parameter set when the following threshold criteria is satisfied,

$$|l(\theta^{old}) - l(\theta^{new})| < \varepsilon, \quad (16)$$

where l is the log-likelihood function and the threshold, ε , is decided. The standard value is a threshold of $\varepsilon = 0.01$, given by a supportive Python package, `hmmlearn`. The covariance matrix is allowed to take the type, `full`, meaning for a given state i , Σ_i computes all entries, holding the variance of each feature

along the diagonal entries and the covariance between feature pairs along the non-diagonal entries; the most flexible option for covariance computation. For small sample sizes, there can be potential risk of overfitting; however, this can be validated with out-of-sample testing and the comparison of statistics like AIC and BIC can prevent the choosing of an 'overfit' model.

4.3 Model Selection

Before model estimation, each feature set was standardised using the training-sample mean and standard deviation. For feature (j), the standardised observation was calculated as

$$z_{t,j} = \frac{o_{t,j} - \mu_{j,\text{train}}}{\sigma_{j,\text{train}}}, \quad (17)$$

where $\mu_{j,\text{train}}$ and $\sigma_{j,\text{train}}$ denote the mean and standard deviation estimated from the training period. Standardisation placed variables measured on different numerical scales onto a comparable basis, preventing features with larger magnitudes from disproportionately influencing estimation of the Gaussian emission distributions. The scaler was fitted exclusively to the training data, and the resulting parameters were subsequently applied to the test sample, thereby avoiding out-of-sample information leakage.

Training a range of candidate models is necessary to compare different model configurations and help identify the combination of hidden states that captures market behaviour. Candidate models were trained using multiple feature vectors in order to assess the contribution of different engineered variables to regime identification. Six models were trained using data spanning the start of 2024 to the end of 2025, with the test sample holding the first 6 months of 2026 data. The first two models (M1 and M2) were allowed to assume a 3 state and 4 state volatility regime framework respectively, trained using only three features: log returns, a volatility ratio, $\frac{\sigma_{10}}{\sigma_{60}}$, and the rolling daily volatility across a 20 day window. This feature vector holds a simple but broad picture, with log returns offering a simple indicator for price action, rolling daily volatility holding a measure of volatility progression and the volatility ratio looking at this progression on a larger scale. Two models (M3 and M4) were also trained using a feature vector of all seven features mentioned in Section 3.3, using again 3 state and 4 state frameworks. Finally we also trained two models (M5 and M6) similarly to M3 and M4, but excluded the volatility ratio feature, as it provides no indication of volatility direction and can potentially cause the identification of regimes changes to lag, itself already being partially correlated to rolling realised volatility anyway.

Following this, trained models were compared using statistical metrics, AIC, BIC and log-likelihood scores. Employing these metrics helps balance statistical goodness of fit and model complexity, reducing the chance we select an overfitted model. These metrics can also identify the least effective features if one wished to train and score all possible combinations of feature vectors. Models with lower AIC and BIC values are desirable alongside a higher log likelihood score, with these metrics displayed below in Table 2. Additionally, statistical performance alone is not sufficient for model selection, leading us to pick model 4 (M4) for which a 4 state model is assumed and the entire feature vector is used. We also require the inferred hidden states correspond to economically interpretable environments, meaning we must reject more statistically favourable models due to a lack of interpretation, such as model 6 (M6).

Furthermore, M4 was subject to further analysis using out-of-sample validation and regime persistence as discussed in the following sections.

Table 2: Information criteria and log-likelihood values for the trained HMM specifications.

Model	Log-Likelihood	AIC	BIC
Model 1	−1339.80	2685.60	2698.49
Model 2	−1293.15	2592.29	2605.19
Model 3	−2964.89	5935.79	5948.68
Model 4	−2692.27	5390.54	5403.43
Model 5	−448.92	903.84	916.74
Model 6	−276.27	558.53	571.43

Moreover, model 4 was selected because it achieved the best information-criterion performance within its comparable specification pair while also producing the most economically meaningful and out-of-sample-stable regime structure. As previously discussed, the comparable models are in pairs, M1 & M2, M3 & M4, M5 & M6. M4 did not produce the numerically lowest information criteria among the four-state specifications. However, direct comparison of the absolute AIC and BIC values is limited because the models were estimated using different feature sets. Within its corresponding feature specification, M4 was strongly preferred to the three-state alternative. It was ultimately selected because it generated the clearest economically interpretable regimes and demonstrated satisfactory persistence and stability out of sample. Table 2 also illustrates a statistical preference for a 4-state model rather than a 3 state model, as tested using M1, M3 and M5, all which underperformed relative to their 4-state alternative.

4.4 Regime Inference

Once the HMM parameters have been estimated, the remaining objective is to infer the latent market regime associated with each observation. Since the regimes are not directly observable, statistical inference must be used to estimate the most probable hidden state sequence consistent with the observed feature vectors and the trained model parameters.

The transition matrix encodes information about the persistence of regime duration from one trading day to the next, however the HMM requires to know the most likely sequence of hidden states over the entire sample. The Viterbi algorithm, as previously outlined in Section 2.1, can be relied upon as it computes the most likelihood state given the current observation vector and the use of posterior probabilities. The Viterbi algorithm is elegant in its method to solve the optimisation problem, $\hat{x} = \operatorname{argmax} P(o|x)$; brute force is intractable given the permutations of hidden states sequence across large time periods, yet Viterbi remembers only the best sequence ending in each state and discards all others. Computationally, the algorithm has complexity $O(N^2T)$, rather than exponential complexity, T here being number of observations and N being the number of hidden states. The algorithm stores a probability $\delta_t(j)$, which gives the probability of the best path ending in state j on day t . Initially, we define $\delta_0(j) = \pi_j b_j(S_1)$ as the path contains the starting observation and the algorithm builds on this recursively,

$$\delta_t(j) = b_j(o_t) \max_i [\delta_{t-1}(i) a_{ij}]. \quad (18)$$

The recursion step starts with every possible previous state i , and takes the probability of best path encoded in $\delta_{t-1}(i)$, multiplying it by the transition probability of state i to j . The max function keeps only the largest of the probability values and multiplies by the emission probability to measure how well the current observation fits state j .

Additionally, methods such as filtering and smoothing can help improve regime interpretability. Filtering is a useful tool for live trading extensions of HMM, helping us consider what current state we believe the market to be in given all observations. On the other hand, smoothing looks at all observations, past and future, helping us consider what state we were in on a given day based on the observations prior and post day; a strong tool for accurate regime inference and analysis of regime characteristics.

4.5 Out-of-Sample Validation

Out of sample testing was used to assess whether the inferred regime structure generalises to previously unseen test data. As the test data was selected chronologically to the train data as mentioned beforehand, the time-series structure of the observations is preserved, thereby preventing look-forward bias. After training, the model parameters were held fixed and the test observations were passed through the fitted HMM to infer their most likely regimes. The generalisation of the model beyond the training sample provides evidence the model has not simply overfit the training data. The statistical characteristics of each inferred regime, including returns, realised volatility, drawdown and VIX levels, were subsequently compared between the training and test samples to assess the consistency of the decoded market states.

Out-of-sample regimes were decoded using the Viterbi algorithm over the complete test sequence. Consequently, the state assigned to a given test observation may depend on observations occurring later within the test period. The resulting sequence should therefore be interpreted as a retrospectively decoded out-of-sample regime path rather than a strictly real-time classification. Although model estimation and feature scaling remained free from test-sample leakage, the timing and apparent stability of individual regime transitions may benefit from this within-test look-ahead effect.

5 Results

The following chapter presents the results obtained from applying the Gaussian Hidden Markov Model using the engineered volatility feature set and Model 4, as described in the previous chapters. The objective of the analysis is to determine whether statistically distinct volatility regimes can be identified, and whether these inferred regimes remain consistent when applied to previously unseen market data.

The results are presented in the order of the modelling pipeline. Initially, the statistical performance of the candidate models was compared using the log-likelihood, Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) in order to justify the selection of the final model. The inferred hidden states were then examined through their statistical characteristics, average duration and transition behaviour, allowing each regime to be assigned an economic interpretation. Finally, the selected model was evaluated on the out-of-sample test period to assess the stability and generalisability of the inferred regime structure beyond the training data.

5.1 Identified Regimes

The chosen candidate model, M4, assumes a 4 state market framework to decode volatility regimes, with the following regimes decoded from statistical information about the volatility-related features used in the feature vector. The model identified 4 distinct regimes during the train sample; a low volatility bull state, (0), with positive mean, very small mean drawdown and the lowest VIX level of all regimes. M4 also proposed a bull momentum state, (1), with characteristics similarly reflecting the low volatility bull state but with larger excess kurtosis, a higher mean VIX level and downside asymmetric skew. Furthermore, we decoded a high volatility state, (2), with the highest VIX level and the largest minimum drawdown. This state doesn't seem to favour either positive or negative returns and primarily reflects periods of rapid rolling volatility increase, as demonstrating in the following Figures 1 and 2. Finally, we have a fourth state, (3), which was identified as a heavy tail-risk regime, due to its extremely high excess kurtosis value and large downside dominated skewness.

Figure 1 illustrates a strong overall upward trend in TSM stock over the two-year training period, interrupted by several sharp but comparatively brief downturns. The low-volatility bull regime, state 0, occupied 49.8% of the sample, while the higher-momentum bullish regime, state 1, accounted for a further 27.6%, highlighting the predominance of positive market conditions. The high-volatility stress regime, state 2, occupied 13.1% of the sample, with the heavy tail-risk regime, state 3, accounting for the remaining 9.5%.

Many of the largest price and volatility movements occurred during periods of heightened uncertainty surrounding AI demand, the semiconductor industry and broader macroeconomic conditions. TSMC's central position in the advanced-semiconductor supply chain, including its manufacturing relationship with NVIDIA and other major customers, leaves its valuation sensitive to changes in industry expectations and customer demand. Such periods of heightened uncertainty are reflected in states 2 and 3: state 2 captures broader high-volatility and deep-drawdown conditions, whereas state 3 appears to represent more idiosyncratic heavy-tail events, as Figure 2 and 3 illustrate.

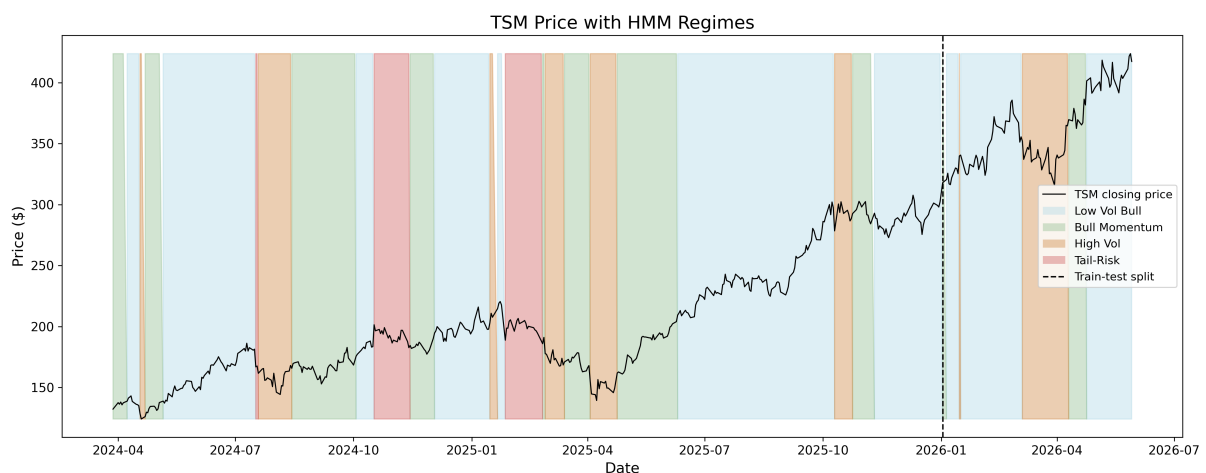


Figure 1: TSM closing price coloured by the inferred hidden volatility regimes. Both the training and OOS tested paths are plotted, separated by a black vertical line.

Comparatively between the two price plots, we see the overall strong upward trend maintaining with high volatility regimes separating bullish regimes. Interestingly, for the test plot, Figure 1, the downside tail-risk regime was not identified throughout the train sample. Without it's appearance across the test sample it is hard to assess the persistence of this regime out-of-sample and therefore it may only be economically meaningful across the train data.

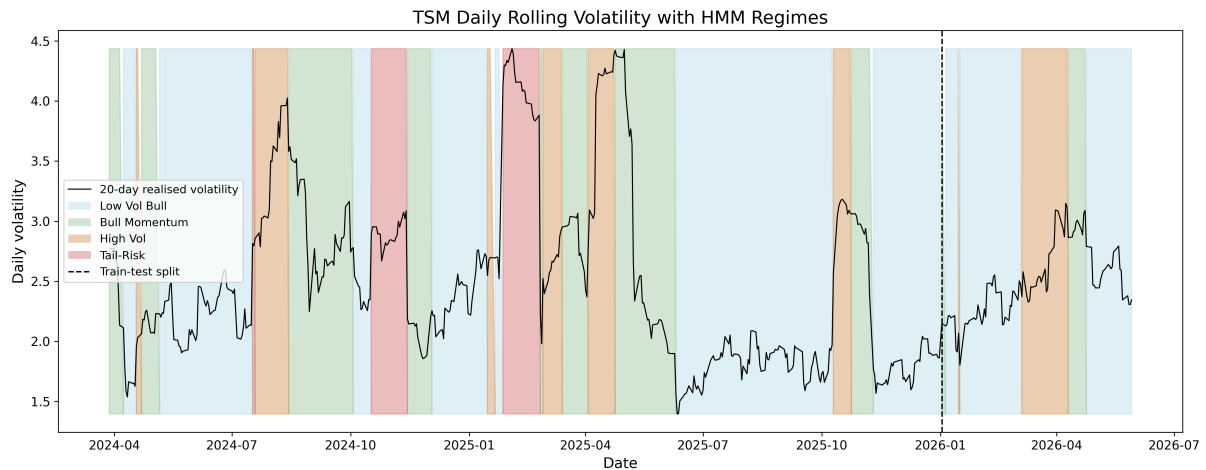


Figure 2: A combined train and test split visual to compare the behaviour of rolling daily volatility both in and out-of-sample.

Figure 2 demonstrates regime identification against rolling volatility, showing how the low volatility bull states exist when volatility lies in the lower volatility ranges, as expected. Similarly, Figure 2 reveals that the high volatility states and the heavy tail-risk states both share an identifiable rapid volatility increase. As a result, we can look towards other metrics contained within the feature vector to further distinguish the economic interpretability. High momentum bull states appear to exist with elevated volatility levels in comparison to the low volatility bull states, with a possible risk premium lying in these regimes due to uncertainty introduced with price rallies. Figure 2 also reveals how sharp volatility increases occur at the beginning of the tail-risk states, with a loosely symmetric sharp decrease ending the regime. This potentially implies that the existence of these states reflect independent macroeconomic events occurring at the time, which introduce short-term elevated uncertainty and a large influx of trading volume.

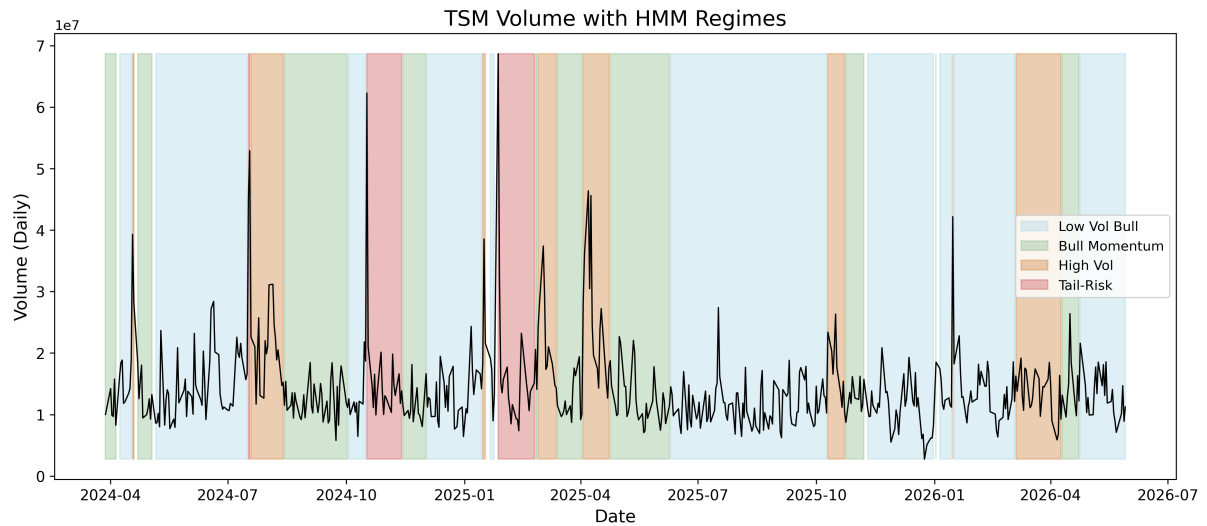


Figure 3: TSM daily trading volume plotted across the full sample with regime coloured background to help visualise the effect of identified regimes on volume changes or influx.

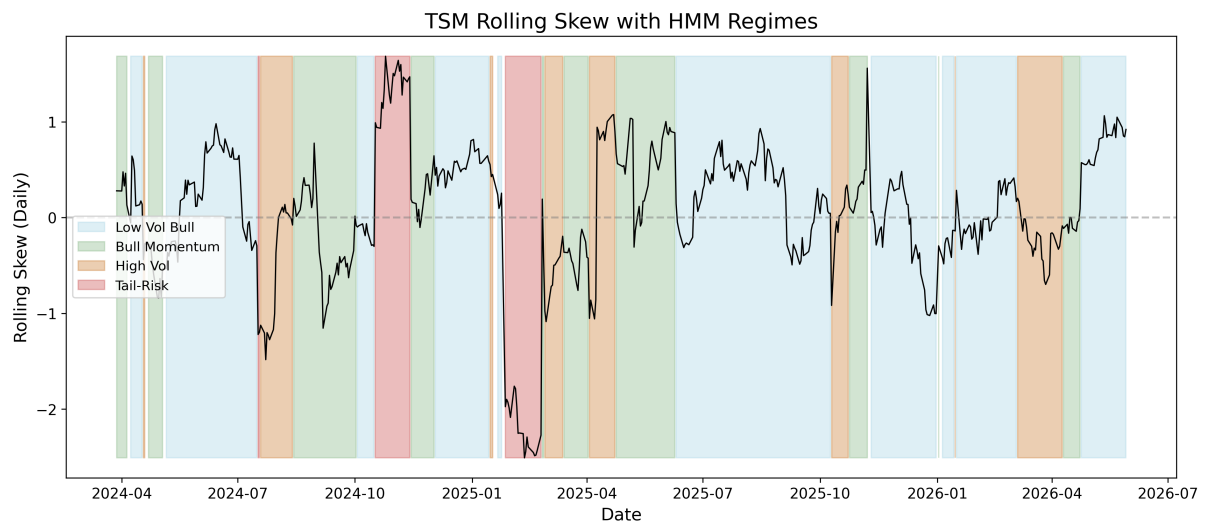


Figure 4: TSM daily rolling skew on a 20-day window. Plotted across the full sample with regime coloured background to illustrate the change in skewness behaviour with regime switches.

Figure 3 shows pronounced increases in daily trading volume around the onset of several downside tail-risk episodes, with some of the largest volume spikes occurring near the transition into these states. Trading volume was not included as an input feature when estimating the Hidden Markov Model, yet it also appears elevated during the high-volatility regimes. This provides useful external support for the economic interpretation of the decoded states and is consistent with the commonly observed positive association between trading activity and return volatility.

The relationship should not necessarily be interpreted as directly causal. Elevated volume and volatility may arise simultaneously in response to new information, investor disagreement, changes in market liquidity or rapid portfolio repositioning. Greater order flow can contribute to larger price adjustments

when market depth is limited, but high trading volume alone does not imply that a large price movement must occur.

By comparison, the daily trading volumes observed in the low-volatility bull and higher-momentum bullish regimes appear visually similar. This suggests that their primary distinction is more likely to arise from differences in return behaviour, realised volatility and the broader market conditions captured by the model, rather than from trading volume alone.

Figure 4 offers a supportive view to separate the high volatility and heavy-tailed states. Comparatively across all states, the tail-risk states as illustrated exhibit much larger skewness, a trait that the high volatility states do not replicate. For the high volatility states, the skewness is not significantly different from the two bullish states. Furthermore, the two identified heavy-tailed states hold positive and negative skewness respectively, suggesting with uncertainty that the regime holds no favouritism to directional skewness, however, additional state observations are required to explore and confirm this more confidently.

5.2 Regime Characteristics

The regimes themselves were analysed and decoded by drawing statistical characteristics from regime summaries for metrics such as log returns, VIX level and drawdown. The regime characteristics were used to help us develop examinable ideas about the regimes, explored with future testing and analysis.

Table 3: Train summary statistics of the inferred hidden regimes.

State	Log Returns				Mean VIX	Drawdown		
	Mean	Std. Dev.	Skewness	Kurtosis	Mean	Mean	Min	Max
0	0.386	2.005	0.1870	-0.0788	15.9512	-0.0282	-0.1046	0.0000
1	0.385	2.117	-0.1435	0.7923	17.9953	-0.1221	-0.2697	0.0000
2	-0.461	3.993	0.4476	0.3407	24.4481	-0.1776	-0.3682	0.0000
3	-0.548	3.612	-1.0176	5.3393	17.3267	-0.07112	-0.1468	0.0000

The rows follow the same order as the state labels in the figures; Table 2 sets out the regime summaries for M4 in training, its data taken from all of 2024-2025 end.

Table 4: Test summary statistics of the inferred hidden regimes.

State	Log Returns				Mean VIX	Drawdown		
	Mean	Std. Dev.	Skewness	Kurtosis	Mean	Mean	Min	Max
0	0.243	2.271	0.322	0.135	17.814	-0.022	-0.089	0.0000
1	0.876	2.675	0.390	-0.680	18.310	-0.032	-0.060	0.0000
2	0.264	3.066	-0.160	0.321	25.065	-0.112	-0.181	0.0000

Table 4 provides comparative regime summaries for M4 when the unseen test data was passed through the fitted HMM. The four-state model demonstrates reasonable out-of-sample stability, with the low-

volatility bullish and high-volatility stress regimes retaining clear economic interpretations across the training and test samples. The intermediate bullish regime exhibits weaker stability, while the downside tail-risk state is not observed during the test period. Consequently, the model displays good economic robustness across its principal regimes, although temporal persistence must be assessed separately using transition probabilities and decoded regime durations.

I briefly investigated the window sensitivity across the features used to train the HMM. Most of the features as previously described, are computed using 20-day rolling windows. Increasing the window caused the model to over generalise, decreasing the accuracy of its regime identification and returning comparatively unfavourable statistical metrics. Windows of 5, 10, and 15 day were also examined but these also mainly struggled to maintain economic interpretability across the out-of-sample testing, when the train-test regime characteristics were analysed. Decoded regimes were over-specific to daily changes and the increase in state-swapping probabilities was invaluable as average state duration sharply decreased.

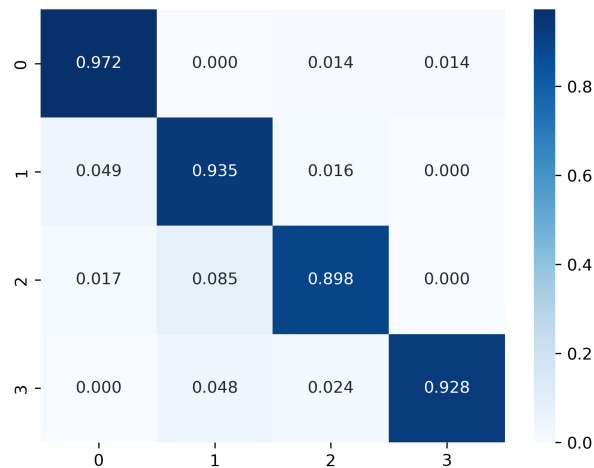


Figure 5: The model transition matrix heatmap evaluated on the training data and showing the probabilities of regime-switching between identified states.

Importantly, an analysis of the transition matrices evaluated by the model must be carried out, as the matrix encodes information about regime persistence, expected duration and the probability of state-switching. Figure 5 demonstrates strong regime persistence for both the low volatility bull state and the high momentum bullish state along the diagonal entries, with low probabilities on the off-diagonal entries indicating a lack of regime-switching intraday. Significantly, the probabilities along a given row add to 1, satisfying a natural probability constraint; the values in Figure 5 have been rounded to three decimal places, yet the true values do sum to unity. The transition matrix holds sensible probabilities such as a completely negligible chance of regime-switching from state 1 to state 3, which is switching from a high momentum state to a tail-risk state. The most likely regime switch is from state 2 to state 1 with probability 8.5% - this being a transition from high volatility state to high momentum state. The state persistence here consolidates the visual evidence proposed by Figure 1.

Overall regime persistence is strong, with main state duration existing with the two bullish states.

Considering the values displayed in Figure 4, the expected duration of each regime can be computed using

$$\mathbf{E}[D] = \frac{1}{1 - A_{ii}}, \quad (19)$$

assuming the state in question is state i . The regime with largest expected duration is state 0, the low volatility bull state, at 35.7 days. Following this is state 1, the high momentum state with an expected duration of 15.4 days. Furthermore, the expected periods of duration were calculated for both the high volatility state and tail-risk state, giving 9.8 days and 13.9 days respectively. The temporal persistence of each regime can be quantified by analysing the empirical Viterbi transition matrix as shown in Figure 6.

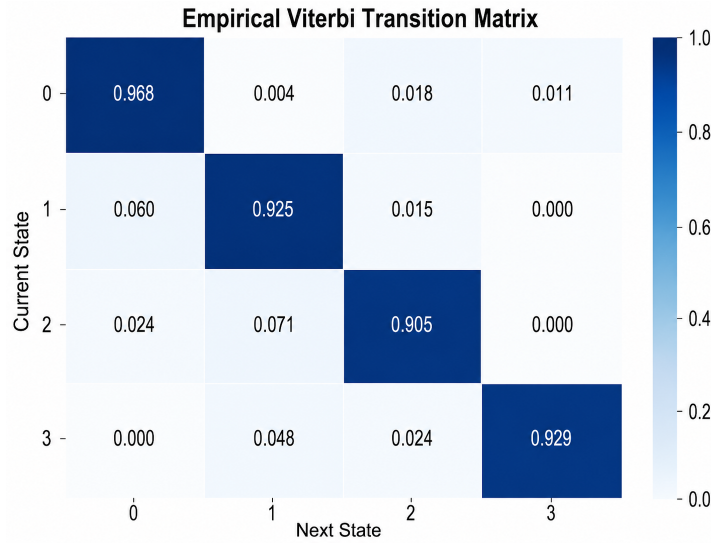


Figure 6: The empirical transition matrix heatmap developed by the Viterbi algorithm.

The empirical transition matrix was constructed by counting transitions between successive hidden states in the Viterbi decoded state sequence and normalising each row to obtain transition probabilities. The empirical matrix differs from the model matrix as illustrated by Figure 5, which represents the realised transition frequencies implied by the most likely sequence of hidden states whereas the model matrix in Figure 6 illustrates a matrix estimated on Baum-Welch optimisation; hence resulting in non-perfect agreement of the matrices yet similar transition probabilities. Both transition matrices are very similar and consolidate that the Hidden Markov Model is temporally persistent. How similar these matrices are can be quantified using the Frobenius norm of their difference, computed as

$$\|A - \hat{A}\|_F = \sqrt{\sum_{i,j} (A_{ij} - \hat{A}_{ij})^2}, \quad (20)$$

where A is the model matrix and $\hat{A}$ is the empirical matrix. Computing the norm gives a value of 0.0240 to three significant figures. The relatively small Frobenius norm indicates limited divergence between the model-implied and empirical transition matrices, providing evidence that the estimated transition dynamics are broadly reflected in the decoded state sequence.

Further regime analysis is provided by examining the conditional distribution of log returns within

each decoded state, complementing the skewness and kurtosis statistics reported in Table 3. Figure 7 reinforces the economic interpretation of the inferred regimes by revealing clear differences in the location, dispersion and tail behaviour of their return distributions. The high-volatility regime displays the broadest distribution and a substantially lower peak, reflecting its greater return variance. Its density extends into both tails and exhibits moderate positive skewness, consistent with the occurrence of large positive rebound returns alongside periods of market stress.

The heavy tail-risk regime is distinguished by pronounced negative skewness and a considerably extended left tail, reaching towards log returns of approximately -0.20. Its high excess kurtosis indicates a greater concentration of extreme observations than would be expected under Gaussian distribution, particularly on the downside. This supports its interpretation as a regime associated with infrequent but severe negative price movements.

By contrast, the two bullish regimes are more tightly concentrated around small positive returns and display substantially lower dispersion. The low-volatility bull regime is approximately symmetric and has near-zero excess kurtosis, making it the closest of the four states to a Gaussian return distribution. The bull-momentum regime is similarly concentrated, although its modest negative skewness and positive excess kurtosis indicate some remaining asymmetry and tail risk. Overall, the distributions support a clear distinction between the relatively stable bullish states and the wider, more tail-sensitive high-volatility and downside-risk regimes.

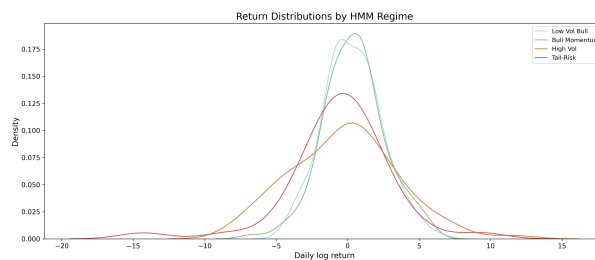


Figure 7: The returns distribution of each state inferred by the Hidden Markov Model for the entire data sample.

5.3 Train-Test Comparison

Considering the statistical evidence alongside the regime labelled price, volatility and return-distribution plots, the selected HMM remains economically interpretable out of sample, although the strength of persistence differs across states. The low volatility bull regime shows the clearest stability. Its daily return volatility rises only moderately from 2.005% in the training sample to 2.271% in the test sample, while mean drawdown remains shallow, changing from -2.82% to -2.20%. The price plot also shows that this regime continues to coincide with sustained upward movements in TSM, particularly during much of the 2026 test period. The bull momentum regime remains associated with positive returns and comparatively moderate volatility, although its out-of-sample characteristics are less stable than those of the low volatility bull state. Mean return increases from 0.385% to 0.876%, volatility rises from 2.117% to 2.675%, and the mean drawdown becomes considerably shallower, moving from -12.21% to -3.20%. This suggests that the test-period observations assigned to bull momentum reflect a stronger and less

drawdown intensive form of upward market movement than in the training sample.

The high volatility regime also remains economically distinct. It retains the highest VIX level in both samples, increasing from 24.448 in training to 25.065 in testing, while realised return volatility remains elevated relative to the two bullish regimes, despite declining from 3.993% to 3.066%. The price and rolling volatility figures support this interpretation, as high volatility episodes occur around periods of heightened volatility and price weakness, including the pronounced decline during early 2026. However, its drawdown severity is lower in the test sample, with mean drawdown moving from -17.76% to -11.20% and the minimum drawdown improving from -36.82% to -18.10%. The tail-risk regime is the only state not identified during the out-of-sample period. Its training-sample characteristics remain economically distinctive negative mean returns, strongly negative skewness and very high kurtosis; but its absence from the test sample means that its persistence cannot be assessed directly. Overall, three of the four regimes remain identifiable and economically meaningful out of sample, with the strongest persistence observed in the low volatility bull and high volatility states. The results therefore provide evidence that the inferred regime structure generalises beyond the estimation sample, while also indicating that the bull momentum state changes somewhat in character and that the tail-risk state is too infrequent for robust out-of-sample validation.

6 Conditional Volatility Estimation

After identifying economically distinct and persistent volatility regimes, the next stage was to assess whether this structure can improve conditional estimates of future volatility. A simple regime-average approach provides a useful benchmark but assigns a constant volatility level to every observation within the same state, ignoring changes in recent shocks and conditional variance. This section therefore evaluates the potential for a regime-dependent forecasting framework, using both an EWMA model and a GARCH(1,1) model to forecast volatility entirely and then using state-dependent information to create comparative forecasts. EWMA as a recursive volatility model estimates conditional variance by assigning greater weight to recent squared returns and progressively smaller weights to older observations. This approach does not hold a fix rolling window as past information decays geometrically over time, instead of being dropped out of the equation, which proves helpful for particular regimes where the volatility shocks exert longer half lives or for faster decaying volatility regimes on the contrary. The conditional variance is given by

$$h_t = \lambda h_{t-1} + (1 - \lambda) r_{t-1}^2, \quad (21)$$

where h_t represents the conditional variance for a time interval t , r_{t-1} representing the previous return and $\lambda \in (0, 1)$ is the decay parameter. This model is adequate for this report as the simplicity of only estimating one parameter compliments the lack of data available for less frequent regimes such as the tail-risk state or the high volatility state. The parameter helps control the balance between historical persistence and responsiveness to new information, with the weight assigned to previous variance given by just the λ value, and the weight on the latest squared return being $1 - \lambda$. Higher values of lambda imply strong volatility persistence and a gradual decay of past information, an intuition about financial markets that has given rise to many other volatility forecasting techniques.

A GARCH(1,1) model was used to estimate the time-varying conditional variance of TSM returns. The model specifies current variance as a function of a constant baseline, the preceding squared return innovation and the preceding conditional variance. The GARCH(1,1) model proposes that the conditional variance is given by

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (22)$$

where α measures the reaction to new shocks, β encodes volatility persistence and ω provides a baseline variance value. The value held by ε represents the unexpected return innovation, given returns are described as $r_t = \mu + \varepsilon_t$ where these innovations follow $\varepsilon_t = \sigma_t z_t$. The sigma term is the conditional variance and z_t is a standardised innovation, $z_t \sim N(0, 1)$. The long-run variance can be given by the expression

$$\frac{\omega}{1 - \alpha - \beta}, \quad (23)$$

provided the constraint $\alpha + \beta < 1$ is satisfied; a constraint that was applied manually alongside the standard constraints, $\alpha, \beta \geq 0$.

The conditional volatility estimates were compared with the corresponding realised-volatility proxy using root mean squared error (RMSE), mean absolute error (MAE) and quasi-likelihood loss (QLIKE). For an estimated conditional variance $\hat{\sigma}_t^2$ and realised variance proxy σ_{20}^2 , QLIKE was calculated as

$$\frac{1}{T} \sum_{t=1}^T \left(\frac{RV_t}{\hat{\sigma}_t^2} + \log \hat{\sigma}_t^2 \right). \quad (24)$$

Lower metric values indicate closer agreement between the conditional variance estimate and the realised variance proxy. QLIKE is useful because it evaluates relative variance errors and penalises substantial underestimation of realised volatility more strongly than equivalent overestimation. RMSE, MAE and QLIKE were therefore considered jointly when comparing the general and state-dependent conditional volatility specifications.

6.1 Regime-Conditioned Volatility Modelling - EWMA

The in-sample results show that the state-dependent EWMA generally follows the broad movements in 20-day realised volatility, while producing a smoother conditional-volatility path. During the low-volatility bull regime, the EWMA remains relatively stable and changes only gradually, consistent with a high decay parameter and persistent volatility dynamics. In contrast, the model responds more sharply during the high-volatility and heavy-tail risk regimes, particularly around the volatility episodes in mid-2024 and early 2025. However, several of these responses exceed the corresponding movements in realised volatility, suggesting that large daily return shocks can temporarily cause the EWMA estimate to overshoot the smoother 20-day measure. This difference is partly expected because the EWMA reacts directly to the most recent squared return, whereas realised volatility is calculated over a rolling 20-day window.

The out-of-sample results provide some evidence that these regime-dependent dynamics remain informative beyond the estimation period. The fixed training-sample decay parameters continue to distinguish

between relatively stable volatility during the low-volatility bull state and faster adjustment during the high-volatility state. In March and early April 2026, the EWMA reacts strongly to the increase in market volatility and captures the broad direction of the rise in realised volatility. Nevertheless, its path is considerably more variable and contains several pronounced overshoots, while the 20-day realised-volatility series responds more gradually because of its rolling-window construction. Following the transition into the bull momentum regime, the two measures converge more closely, although the EWMA subsequently remains above realised volatility during the later low volatility period. Overall, the out-of-sample graph suggests that the model preserves the intended state-dependent responsiveness, but also indicates a tendency to overestimate volatility after large shocks and to decay towards lower volatility more slowly than the realised measure.

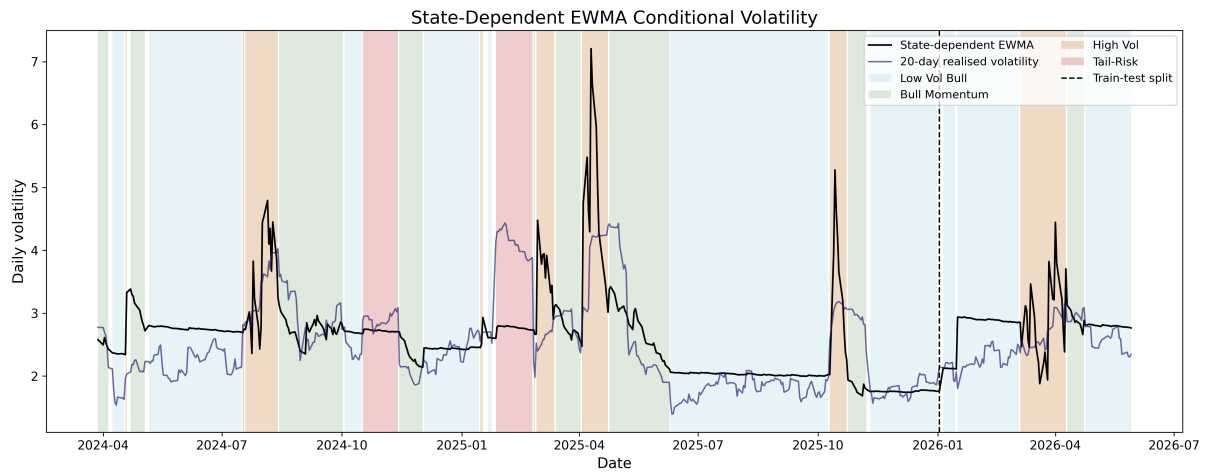


Figure 8: State-dependent EWMA volatility forecasted against the 20-day rolling volatility feature. The training path is shown along with the OOS tested path, separated by a vertical black line.

Quantitatively, the λ value for the high volatility state was estimated to be 0.711, implying a shock weight of 0.289. Compared to the other regime values, which are displayed following, this highlights how state-dependent forecast can adapt to regime specific shock persistence and help capture shock amplification difference across our market states. The values for the low volatility bull state, the high momentum state and the heavy-tail risk state were, 0.995, 0.935 and 0.995 respectively; much larger volatility persistence and slow adjustment to new shocks. The low shock responsiveness in the bullish states are not that surprising, given the volatility characteristics for these regimes were the lowest of all 4 states during state decoding. However, the slow adjustment to shocks in both the tail-risk states as indicated by the EWMA may appear interesting, following our suggestion of macroeconomic events influence on TSMC and massive volume influx. Figure 8 shows how the rolling 20-day volatility captures a large volatility move in February 2025, which the EWMA forecast negates completely and fails to capture. This move is instead portrayed by the EWMA as being in the high volatility state which follows, keeping consistent with regime interpretability.

Comparing statistical measures for the out-of-sample test on the EWMA forecast can also help to determine how accurate the forecast is with respect to the observation data available. Given both measures are

quantified graphically in percentage points, the MAE was reported as 0.393, and the RMSE was reported as 0.487 to 3dp. Out-of-sample evaluation showed that the standard EWMA benchmark outperformed the state-dependent specification across all three loss functions. The general EWMA produced an RMSE of 0.253 and an MAE of 0.195 percentage points, compared with 0.487 and 0.393 respectively for the regime-dependent model. QLIKE also marginally favoured the general specification. This indicates that the additional regime-specific decay parameters did not provide incremental forecasting value over the test period. Although the state-dependent model responded more aggressively during stressed regimes, this frequently resulted in volatility overshooting, while the common-lambda model produced a smoother and more accurate forecast path.

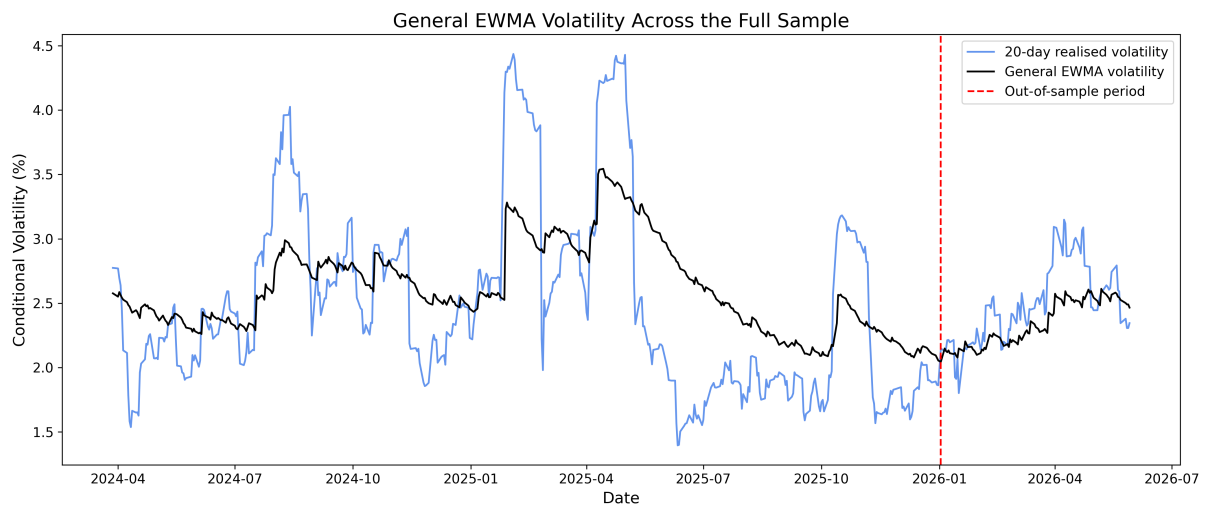


Figure 9: General EWMA volatility forecast across the entire data sample, train and test, compared to the rolling 20-day volatility measure.

Figure 9 shows the general EWMA across the entire dataset, using only one lambda value and ignoring state-dependence. The general EWMA produces a smoother volatility path than the 20-day realised-volatility measure because its fixed decay parameter assigns geometrically declining weights to past squared returns. During clusters of unusually large returns, the rolling 20-day estimator can rise above the EWMA because all observations within the window retain equal weight until they exit the sample. The EWMA instead spreads the impact of shocks through its recursive variance estimate, reducing the magnitude of short-lived volatility peaks but allowing their effects to decay more gradually. This is evident with volatility spikes occurring in February 2025 and May 2025, where the rolling volatility illustrates larger amplification but a rapid decay, compared to a small, less aggressive amplification but a longer, more gradual decay described by the EWMA forecast. Conclusively, the use of EWMA as an example, potentially indicates that state-dependent volatility forecasting has weak predictive power in comparison a general EWMA benchmarks, overshooting volatility spikes and appearing to slow to adjust to changes across other regimes.

6.2 Regime-Conditioned Volatility Modelling - GARCH (1,1)

Persisting the exploration of whether state-dependent frameworks increase the accuracy of conditional volatility estimates, a GARCH(1,1) model was implemented as another comparative example. The

analysis does not attempt to predict the exact future path of realised volatility. Instead, state-dependent EWMA and GARCH specifications are used to estimate a time-varying conditional-variance process and are compared with corresponding state-independent benchmarks. Their estimated volatility paths are evaluated against the contemporaneous 20-day rolling realised-volatility measure in order to assess whether conditioning model parameters on the prevailing HMM state improves the representation of changing volatility environments

Extending the state-dependent GARCH model to the out-of-sample period in Figure 10 reveals an interesting generalisation. Although the regime-conditioned specification continues to produce a smoother volatility path, the red series no longer aligns with the economic interpretation of the high-volatility state. During March and April, both the 20-day realised volatility measure and the Gaussian HMM indicate a sustained high-volatility episode, yet the state-dependent GARCH forecast declines throughout much of this period and only reconverges with the realised and general GARCH paths after the regime ends. Figure 3 also shows that this episode was not accompanied by a pronounced increase in trading volume, unlike several other high-volatility regimes identified in the sample. This creates two possible interpretations: either the HMM has misclassified the episode, or the state-dependent GARCH specification has failed to translate the regime information into a robust out-of-sample volatility forecast.

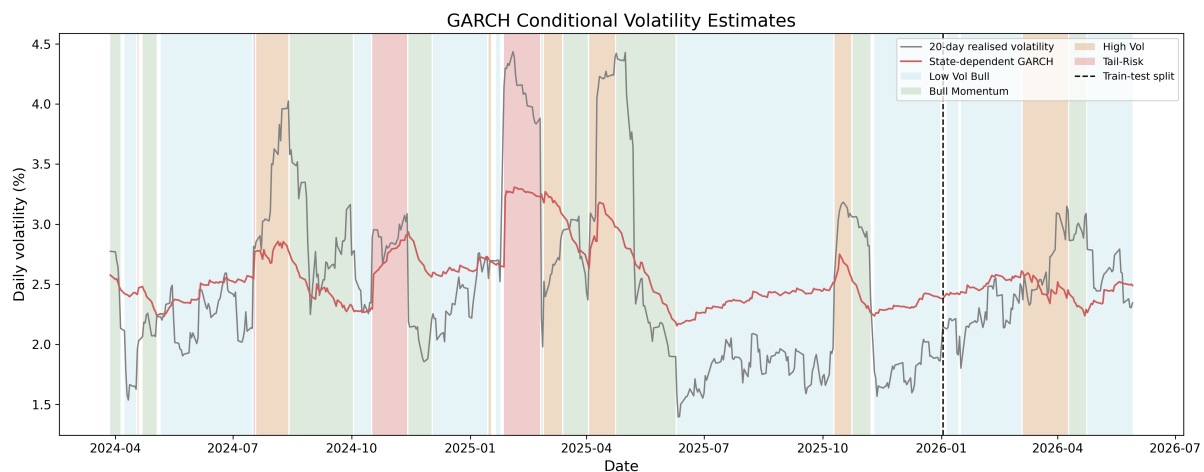


Figure 10: State-dependent GARCH (1,1) volatility path compared against the 20-day rolling volatility benchmark. Both the training and test paths are visualised and separated by the vertical line.

From a quantitative perspective, the state-dependent GARCH specification estimates common α and β parameters across all regimes, while allowing each state to have a distinct variance intercept, ω 's. This pooled structure reduces the number of parameters requiring estimation and limits overfitting when some regimes contain relatively few observations. The estimated coefficients are $\alpha=0.01597$ and $\beta=0.9711$, giving $\alpha + \beta = 0.9871 < 1$. The stationarity condition is therefore satisfied, although the value close to one indicates highly persistent conditional variance. The regime specific intercepts for States 0–3 are estimated as 0.1243, 9.508×10^{-7} , 0.001176, and 0.2380, respectively.

The small value of α suggests that the model responds only weakly to new squared return innovations, while the large value of β implies that previously estimated volatility decays very slowly. This combination helps explain why the model smooths the volatility path and underestimates the magnitude of short-lived

peaks in the training sample. It also produces extended decay tails following local volatility spikes, as shown in Figure 10, because shocks are carried forward primarily through the lagged conditional-variance term.

The ordering of the state-specific intercepts is less economically intuitive. In particular, the high-volatility regime receives a substantially smaller intercept than the low-volatility bull and downside tail-risk states. Since α and β are held constant across regimes, the model cannot compensate through stronger shock sensitivity or persistence within the high-volatility state. This may explain why the regime-conditioned forecast does not always reproduce the expected volatility ordering implied by the HMM.

Some apparent disagreement with the 20-day realised-volatility series may nevertheless reflect the construction of the benchmark rather than a complete failure of the GARCH model. GARCH assumes that volatility shocks decay gradually, consistent with volatility clustering in financial markets. By contrast, a rolling measure can fall abruptly when a large return observation exits the 20-day window. The sharper declines in realised volatility therefore partly reflect the mechanical behaviour of the rolling window, whereas the more gradual GARCH decay reflects persistent conditional variance.

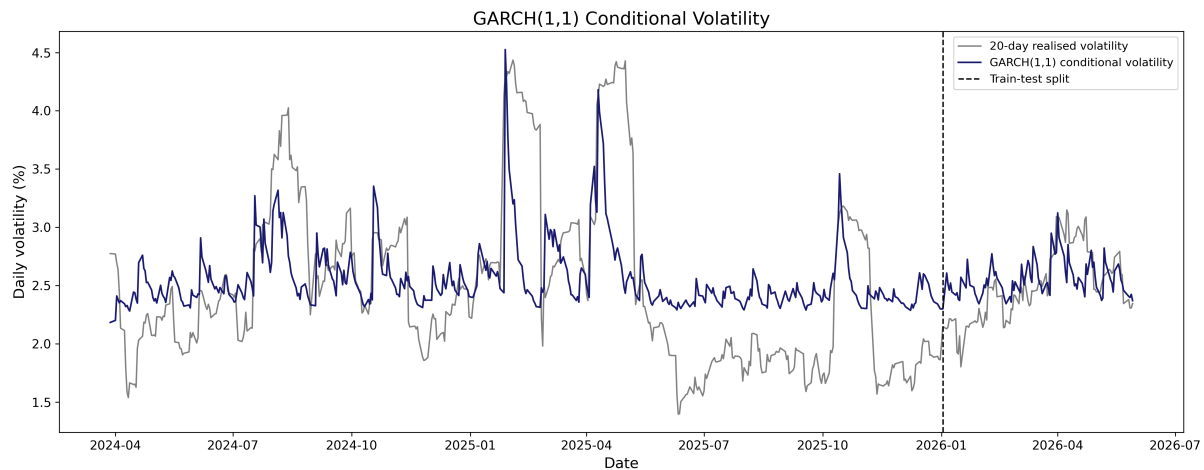


Figure 11: Standard GARCH (1,1) volatility path plot against the rolling 20-day average for the training period of data. The plot is coloured by regime as indicated in the legend.

Figure 11 reinforces the conclusion that introducing state dependence primarily acts as a smoothing mechanism rather than improving forecast accuracy. The general GARCH(1,1) path in blue more frequently captures volatility peaks with amplitudes closer to those observed in the 20-day realised-volatility series. It also appears to preserve the economic interpretation of the regimes more consistently across the training sample, while still providing a stable underlying volatility estimate through periods of noisy returns and isolated shocks.

The quantitative results support this interpretation. The general GARCH model achieved an RMSE of 0.268, compared with 0.344 for the state-dependent specification, and an MAE of 0.215 against 0.283. The QLIKE scores were closer, at 2.835 for the general model and 2.852 for the state-dependent model, although the general specification still performed marginally better. Taken together, these results indicate that the additional regime-dependent intercepts did not provide sufficient forecasting value to offset the

loss of flexibility created by imposing common shock-response and persistence parameters across all states. The state-dependent GARCH remains useful as an interpretative extension, demonstrating how the HMM regimes can be incorporated into a conditional-variance framework, but the standard pooled GARCH(1,1) provides the more robust and accurate volatility forecast in this application.

7 Extensions

The methodology developed throughout this report represents one possible framework for modelling latent volatility regimes from observable market data. However, as with any statistical model, its performance is constrained by the assumptions adopted during model specification and training. These limitations motivate several natural avenues for future development, including improvements to the statistical assumptions of the Hidden Markov Model, more sophisticated representations of regime dynamics, and extensions towards broader financial applications. The following subsections discuss several of these potential developments and their relevance to quantitative volatility modelling.

7.1 Heavy Tailed Emission Distributions

Assuming our emission distributions are Gaussian has been seen a sensible assumption for sensible states, however it fails when financial markets exhibit heavy tails, extreme observations and volatility clustering. A Gaussian distribution assigns relatively low probability to these regions and struggles therefore to capture this phenomena accurately. The crisis state and volatile recovery state both insist that returns no longer follow a Gaussian as the return figures illustrate large excess kurtosis for these states; encouraging us to turn to emission distributions suitable for modelling heavy tails. An effect first extension would be to explore the improvements made when using a student-t distribution to model emissions. The incorporation degrees of freedom, ν , allows the model to more naturally capture extreme events as it controls tail thickness, with the probability density function given as

$$f(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\Gamma(\frac{\nu}{2})} \cdot (1 + \frac{x^2}{\nu})^{-\frac{\nu+1}{2}}, \quad (25)$$

where Γ denotes the gamma function. The framework for the HMM would remain largely unchanged, however, the Baum-Welch algorithm would now have to estimate the number of degrees of freedom, increasing the optimisation difficulty and possibly increasing the chance of overfitting.

7.2 Dynamical Hidden Markov Models

The main limitation of our Gaussian HMM besides its struggle to capture heavy tails is the assumption our transition matrix remains fixed. Though a simplification, this is obviously unrealistic within financial markets as regimes clearly behave different under different macroeconomic conditions. Naturally, extensions towards a state-dependent transition matrix or an introduction of time dependence would improve the model. We can adopt Ntantamis's ideas to incorporate macroeconomic variables within the HMM, applying them instead to model a dynamic transition matrix,

$$P(x_t = j | x_t = i) = f(Z_t), \quad (26)$$

where Z_t encodes information regarding interest rates, yield spread and other indicators. This will help the transition matrix remain consistent with the current macro regime and calculate regime-switching probabilities accordingly. Commonly, logistic models are used as probabilities remain between zero and one, the function being defined a multinomial logistic regression,

$$P_{ij}(t) = \frac{\exp(\beta_{ij}^T Z_t)}{\sum_k \exp(\beta_{ik}^T Z_t)}. \quad (27)$$

Extending the model to become a dynamic HMM would improve crisis detection and provide more accurate regime-switching due to proper macroeconomic interpretation, potentially also improving the forecasting ability of the model. Similarly to our other extension suggestion, a dynamic HMM would have a comparatively harder time optimising than our Gaussian HMM would and there would again exist a likelihood we reduce how generalisable the model is.

One key limitation of the current model is that there doesn't exist a mechanism for recognising that the probability of a regime change should itself increase before a market transition occurs, this being a direct result of constant transition matrix. The implementation of a dynamic HMM would incorporate a macroeconomic picture like discussed, allowing the model to respond proactively to changing market conditions than relying solely on observations being emitted from the hidden states.

7.3 Extended Testing

Given the model's relative simplicity and the limited quantity of available data, the empirical testing remains necessarily constrained. A useful extension would be to apply the same HMM framework to comparable semiconductor equities, such as NVIDIA or Broadcom, and statistically compare the resulting regime characteristics with those reported in Section 5. Cross-asset analysis could reveal whether the identified states reflect broader sector-wide behaviour or are specific to TSM, while also helping to identify potential weaknesses or biases in the model's regime classification.

A broader dataset would also allow a more rigorous comparison of alternative volatility-forecasting methods. Evaluating multiple specifications over the same out-of-sample period would support a more convincing conclusion on whether HMM regime information provides incremental value for short-term volatility forecasting.

8 Conclusion

This section draws together the principal findings of the study and evaluates the extent to which the proposed framework successfully identified and forecast changes in TSM volatility. It first summarises the economic and statistical insights obtained from the Gaussian HMM, EWMA and GARCH specifications, before considering the main limitations affecting the robustness and interpretation of the results. The conclusion then outlines several areas for future development, including a comparison with option-implied volatility and further refinement of the regime-dependent forecasting framework.

8.1 Summary of Findings

The Gaussian Hidden Markov Model successfully identified four economically interpretable volatility regimes in TSM over the two-year training period, with the resulting state structure generalising reasonably well across the seven-month out-of-sample test period. The model distinguished between a low-volatility bull state, a high-momentum bull state, a high-volatility state and a downside tail-risk state using a set of self-engineered features designed to capture changes in return behaviour, realised volatility, market stress and distributional shape. The regime classifications were then incorporated into state-dependent EWMA and GARCH forecasting frameworks and compared with their general counterparts using 20-day realised volatility as the benchmark. Across both approaches, state dependence acted primarily as a smoothing mechanism rather than improving forecast accuracy. The general EWMA and GARCH specifications produced more responsive and economically consistent volatility paths, while the state-dependent models tended to dampen short-lived peaks and, in some periods, failed to preserve the expected volatility ordering of the identified regimes.

8.2 Project Limitations

The principal limitation of this study was the relatively small dataset used for both model estimation and out-of-sample evaluation. The conclusions drawn therefore apply only to the sample considered and should not be interpreted as definitive evidence on the wider forecasting value of regime-dependent volatility models. A larger dataset would improve the reliability of the estimated parameters in the EWMA and GARCH specifications, provide more observations within each regime and allow the model to capture a broader range of market conditions. It would also make step-forward or rolling-window validation more practical, enabling the stability of the state-dependent forecasts to be tested across multiple out-of-sample periods rather than a single test window.

A further limitation is that the forecasting framework was evaluated only on TSM. As proposed in Section 7.3, cross-asset testing would provide a stronger assessment of whether the results are specific to this stock or reflect a more general property of regime-based volatility forecasting. Applying the same feature set and modelling framework to assets with different liquidity, sector exposure and volatility characteristics would allow more robust conclusions to be drawn about regime-identification accuracy and the behaviour of indicators such as realised volatility, skewness and trading volume during regime transitions.

Finally, the Gaussian HMM assumes that each regime's feature distribution is conditionally Gaussian. This may be restrictive for financial data, particularly in the high-volatility and downside tail-risk states, where heavier tails, excess kurtosis and pronounced skewness were observed. Section 7.1 therefore considers the use of Student-t emission distributions, whose additional degrees-of-freedom parameter would allow the model to accommodate extreme observations more naturally. Such an extension may improve regime separation during stressed periods and reduce the risk of forcing heavy-tailed behaviour into an inappropriate Gaussian specification.

8.3 Future Work

The most important extension would be to repeat the analysis using a longer dataset and multiple out-of-sample windows. This would allow the stability of the identified regimes, transition dynamics and forecasting results to be assessed across a wider range of market conditions. The existing training and test-window lengths could be retained within a rolling or expanding-window validation framework, allowing the model to be re-estimated sequentially and evaluated over several distinct test periods. Alternatively, a longer initial training period could be used to improve parameter estimation and provide a larger number of observations within the less frequent high-volatility and downside tail-risk states.

A further extension would be to estimate an HMM with Student-t emission distributions. The additional degrees-of-freedom parameter would allow each state to accommodate heavy-tailed observations more naturally, which may improve the separation of stressed regimes and reduce the influence of isolated extreme returns on state classification. The use of time-varying transition probabilities should also be explored. Rather than assuming a fixed transition matrix throughout the sample, transition probabilities could be modelled as functions of observable variables such as volatility, drawdown, trading volume or implied volatility. This would allow the likelihood of entering or leaving a regime to evolve with market conditions and could provide more informative signals around periods of regime instability. Importantly, the objective would not be for the transition probability to converge to one whenever a switch occurs, but for the model to assign a higher switching probability when the available information indicates that a transition has become more likely.

Finally, a stricter pseudo-real-time evaluation would classify each observation using only information available up to that date. This could be implemented using filtered state probabilities or recursive Viterbi decoding on an expanding test window. Such an approach would remove the look-ahead component introduced by decoding the complete test sequence and would provide a more realistic assessment of regime-identification performance in live use.

These extensions should be evaluated using out-of-sample likelihood, regime-classification stability and volatility-forecast metrics to determine whether the additional flexibility produces genuine predictive gains rather than merely improving in-sample fit.

References

- Bergomi, Lorenzo (2015). *Stochastic Volatility Modeling*. Boca Raton, FL: CRC Press. DOI: 10.1201/b19649.
- Ding, Zhuangxin (2016). *Volatility Modeling Using GARCH: Theory and Practice*. SSRN Working Paper. The Risk Protocol Inc. DOI: 10.2139/ssrn.2359212. URL: <https://ssrn.com/abstract=2359212>.
- Han, Yufeng, Ting Hu, and David A. Lesmond (2015). “Liquidity Biases and the Pricing of Cross-Sectional Idiosyncratic Volatility around the World”. In: *Journal of Financial and Quantitative Analysis* 50.6, pp. 1269–1292. DOI: 10.1017/S0022109015000605.
- Ntantamis, Christos (2010). *A Duration Hidden Markov Model for the Identification of Regimes in Stock Market Returns*. CREATES Research Paper 2010-51. Department of Economics and Business Economics, Aarhus University. DOI: 10.2139/ssrn.1343726. URL: <https://ssrn.com/abstract=1343726>.
- Rabiner, Lawrence R. (1989). “A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition”. In: *Proceedings of the IEEE* 77.2, pp. 257–286. DOI: 10.1109/5.18626.
- Stock Analysis (2026). *Taiwan Semiconductor Manufacturing Company Market Capitalization*. URL: <https://stockanalysis.com/stocks/tsm/market-cap/> (visited on 07/30/2026).
- Taiwan Semiconductor Manufacturing Company Limited (2026). *2025 Annual Report*. Annual Report. Taiwan Semiconductor Manufacturing Company Limited. URL: <https://investor.tsmc.com/static/annualReports/2025/english/index.html> (visited on 07/30/2026).